LESSON 6.13

Prime Factorization

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write a number as a product of its prime factors using a factor tree.

WHY THIS MATTERS The space station received 60 supply crates. Mission Control needs to break this quantity into its prime components so the sorting robots can distribute them into equally sized pods. Help the crew find the prime factorization of 60!

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Prime Factorization *Spanish: Factorización prima*	Writing a whole number as a product of only prime numbers.	36 = 2 × 2 × 3 × 3 = $2^2 \times 3^2$
Prime number *Spanish: Número primo*	A number bigger than 1 that you can only divide by 1 and itself.	7 has only two factors: 1 × 7. So 7 is prime.
Composite number *Spanish: Número compuesto*	A number bigger than 1 that you can divide by more than just 1 and itself.	12 = 1 × 12, 2 × 6, 3 × 4 — six factors, so 12 is composite
Factor *Spanish: Factor*	A factor of a whole number divides it evenly, with no remainder. For example, 3 is a factor of 12 because 12 ÷ 3 = 4.	Factors of 12: 1, 2, 3, 4, 6, 12 — each divides 12 evenly.
Factor tree *Spanish: Árbol de factores*	A picture that splits a number into its prime numbers, step by step.	24 → 4 × 6 → (2 × 2) × (2 × 3) → 2 × 2 × 2 × 3
Exponent *Spanish: Exponente*	A small number that tells how many times to multiply a number by itself.	2^3 means 2 × 2 × 2 = 8

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

I want the prime factorization of 60. I start by splitting it: 60 = 6 × 10.

1. 6 is not prime, so I break it: 6 = 2 × 3. Both 2 and 3 are prime, so I stop those branches.
2. 10 is not prime, so I break it: 10 = 2 × 5. Both 2 and 5 are prime, so I stop.
3. Now every factor is prime: 60 = 2 × 2 × 3 × 5.
4. Using exponents, I write it as $2^2 \times 3 \times 5$. I check: 2 × 2 × 3 × 5 = 60. Correct.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's factor 12 together. What is one way to split 12 into two factors? (You might say 3 × 4.)

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I know _____ is prime because its only factors are _____ and _____. I know _____ is composite because it has factors like _____.

Word bank Prime Factorization | Prime number | Composite number | Factor | Factor tree | Exponent | prime | composite | factors | divisible | two factors | more than two

WATCH OUT!

A common mistake

A common mistake in Prime Factorization is stopping the factor tree too soon

LESSON 6.13

Prime Factorization

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each expression: is it a COMPLETE prime factorization, or does it still have composite factors?

Groups: Complete Prime Factorization | NOT Complete (Has Composite Factors)

_____ 2 × 2 × 3 (for 12)

_____ 2 × 3 × 5 (for 30)

_____ 2 × 2 × 2 × 3 (for 24)

_____ 4 × 3 (for 12)

_____ 6 × 5 (for 30)

_____ 8 × 3 (for 24)

HINT 'Complete' means every single factor shown is prime. Even one composite factor makes the whole expression NOT complete.

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Which of the following is a prime number?

(A) 9 (B) 15

(C) 17 (D) 21

HINT Re-read what 'prime' means: exactly two factors, 1 and itself. Any extra factor makes a number composite.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is the prime factorization of 30?

(A) 2 × 3 × 5
(B) 2 × 15
(C) 3 × 10
(D) 5 × 6

HINT A prime factorization uses ONLY prime numbers. Look at each choice: are ALL of its factors prime?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is the prime factorization of 18?

(A) 2 × 3 × 3
(B) 2 × 9
(C) 3 × 6
(D) 6 × 3

HINT Check every factor in each choice — numbers like 9 and 6 are composite, so they can't appear in a prime factorization.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Which of these numbers is composite?

(A) 23
(B) 27
(C) 29
(D) 31

HINT Composite means MORE than two factors. Ask for each number: can I split it into two smaller whole numbers?

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Find the prime factorization of 20** Start: 20
2. **First split** 20 = 4 × 5
3. **Check each factor** 5 is prime, so that branch is done.
4. **Final answer** 20 = 4 × 5

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** Look at every factor in the final answer — is each one a prime number?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How did you decide whether a number is prime or composite? What strategy did you use?

SENTENCE STARTER I know ___ is prime because its only factors are ___ and ___. I know ___ is composite because it has factors like ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.13

Prime Factorization

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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WHY THIS MATTERS The space station received 60 supply crates. Mission Control needs to break this quantity into its prime components so the sorting robots can distribute them into equally sized pods. Help the crew find the prime factorization of 60!

PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each number to its prime factorization.

_____ 12	(A) 2 × 2 × 3
_____ 18	(B) 2 × 3 × 3
_____ 20	(C) 2 × 2 × 5
_____ 28	(D) 2 × 2 × 7
_____ 45	(E) 3 × 3 × 5
_____ 50	(F) 2 × 5 × 5

2 **Complete the table. Show how you found each value.**

Build the factor tree. Write each number as a product of primes.

Number	First Split	Prime Factorization
24		
36		
40		
56		

SHOW YOUR WORK

3 **Write the name of the correct group on each line.**

Sort each number: can it be expressed as a product of exactly TWO prime factors, or does it need THREE or more?

Groups: **Product of Exactly Two Primes** **Product of Three or More Primes**

_____ 6

_____ 10

_____ 15

_____ 35

_____ 12

_____ 30

_____ 24

_____ 60

4 **Circle the letter of the best answer. Show how you know.**

Two students found different factor trees for 60. Student A started with 2 × 30. Student B started with 6 × 10. Which statement is true?

(A) Only Student A gets the correct prime factorization

(B) Only Student B gets the correct prime factorization

(C) They will get different prime factorizations

(D) Both get the same prime factorization: 2 × 2 × 3 × 5

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Find the prime factorization of 45, using exponents** Start: 45
2. **First split** 45 = 5 × 9
3. **Break down 9** 9 = 3 × 3, so 45 = 5 × 3 × 3
4. **Write with exponents** 45 = 3^2 × 5^2

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How did you decide whether a number is prime or composite? What strategy did you use?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.13

Prime Factorization

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Write the prime factorization using exponents.

Number	Prime Factorization (expanded)	Using Exponents
72		
100		
54		

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find Marcus's Mistake

1. **Find prime factorization of 48** Start: 48
2. **First split** 48 = 6 × 8
3. **Break down 6** 6 = 2 × 3
4. **Break down 8** 8 = 2 × 4
5. **Final answer** 48 = 2 × 3 × 2 × 4

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

Choose 24, 36, or 60. Show two factor trees with different first factor pairs. Finish both trees with prime factors, then explain how you checked that both products equal your chosen number.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How did you decide whether a number is prime or composite? What strategy did you use?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help